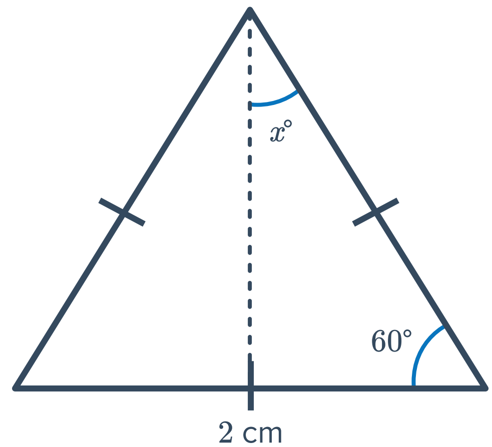
Special right triangles

1 In a $30^\circ - 60^\circ - 90^\circ$ triangle, identify which leg is shorter.

2 Find the exact value of the following:

a $\sin 60^\circ$ b $\cos 60^\circ$ c $\tan 30^\circ$ d $\sin 45^\circ$

3 Consider the following equilateral triangle.
Find the exact values of the following:



a Perpendicular height of the triangle

b Size of angle x

c
- i $\sin 60^\circ$
- ii $\cos 60^\circ$
- iii $\tan 60^\circ$

d
- i $\sin x$
- ii $\cos x$
- iii $\tan x$

Exact value trigonometric ratios

4 Consider $\sin \theta = \dfrac{\sqrt{3}}{2}$.

a Find the value of θ, given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\cos \theta$.

5 Consider $\sin \theta = \dfrac{1}{\sqrt{2}}$.

a Find the value of θ, given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the exact value of $\tan \theta$.

6 Consider $\cos\theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\sin\theta$.

7 Consider $\cos\theta = \frac{1}{2}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\tan\theta$, correct to one decimal place.

8 Consider $\tan\theta = 1$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\sin\theta$, correct to two decimal places.

9 Consider $\tan\theta = \frac{1}{\sqrt{3}}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\sin\theta$.

10 Consider $\tan\theta = \frac{1}{\sqrt{3}}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\cos\theta$, correct to one decimal place.

11 Consider $\cos\theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\tan\theta$, correct to two decimal places.

12 Consider $\sin\theta = \frac{\sqrt{3}}{2}$.

a Find the value of θ, given that $0^\circ \le \theta \le 90^\circ$.

b Now, find the value of $\tan\theta$, correct to two decimal places.

13 Consider $\tan\theta = \sqrt{3}$.

a Find the value of θ, given that $0^\circ \leq \theta \leq 90^\circ$.

b Now, find the value of $\sin 60^\circ$, correct to one decimal place.

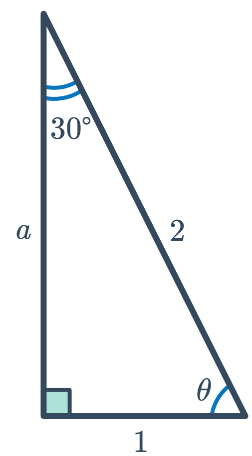
14 For each of the following, find the value of θ given that $0^\circ \leq \theta \leq 90^\circ$:

a $\cos\theta = \dfrac{1}{\sqrt{2}}$ **b** $\tan\theta = \sqrt{3}$ **c** $\sin\theta = \dfrac{\sqrt{3}}{2}$ **d** $\tan\theta = \dfrac{1}{\sqrt{3}}$

15 Consider the following triangle:

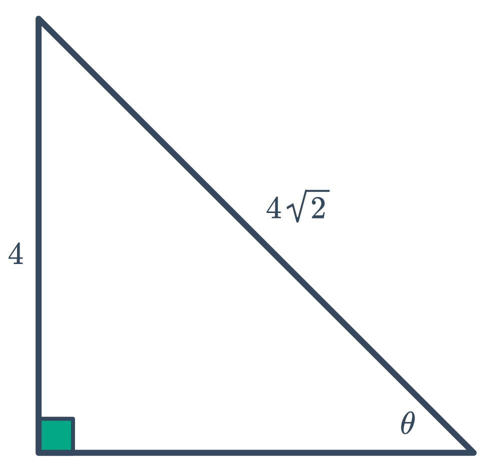
a Find the exact value of a.

b Find the value of θ.

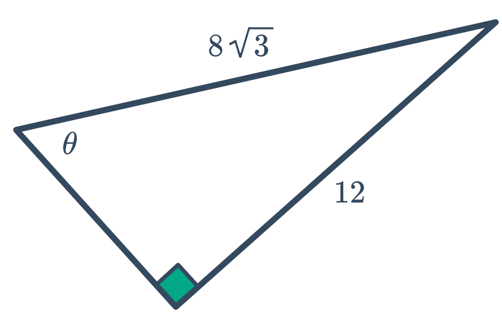


16 Use exact values to solve for θ in the following triangles:

a

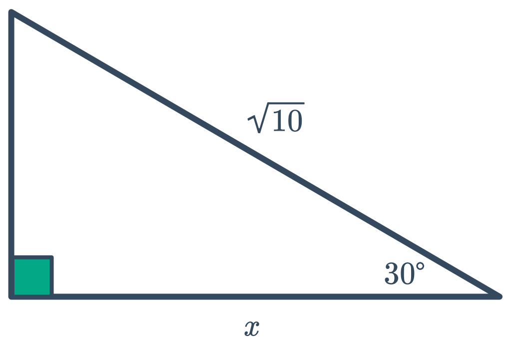


b

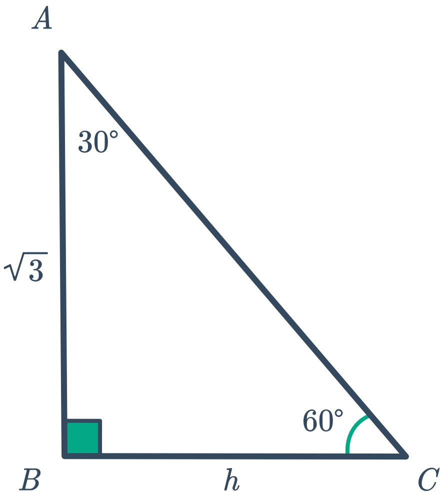


17 Find the exact value of the variable(s) in the [redacted]wing triangles:

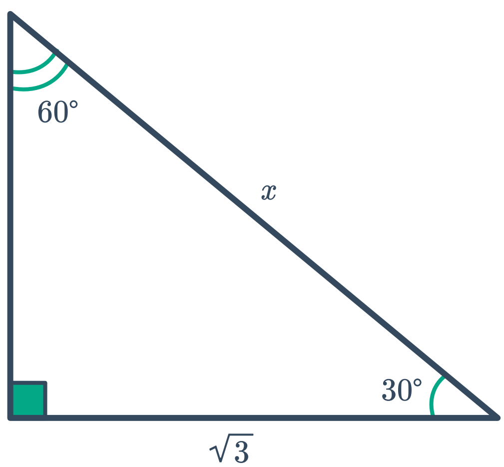
a



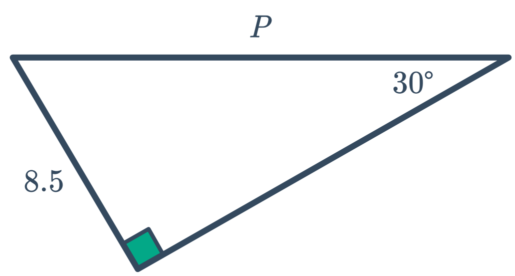
b



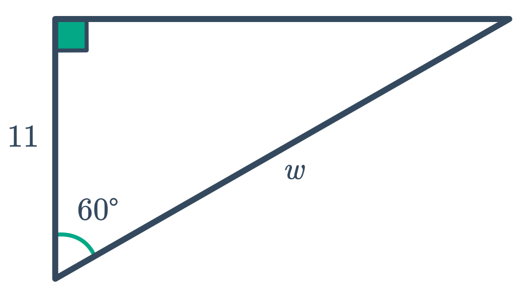
c



d



e



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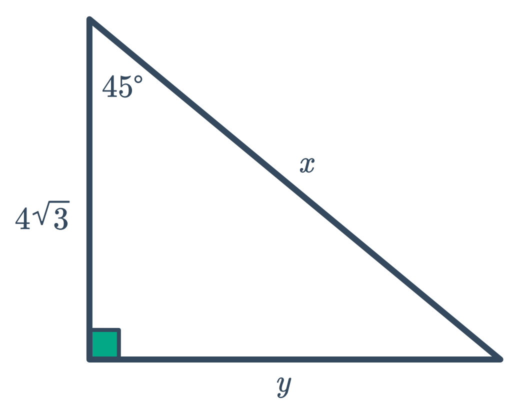
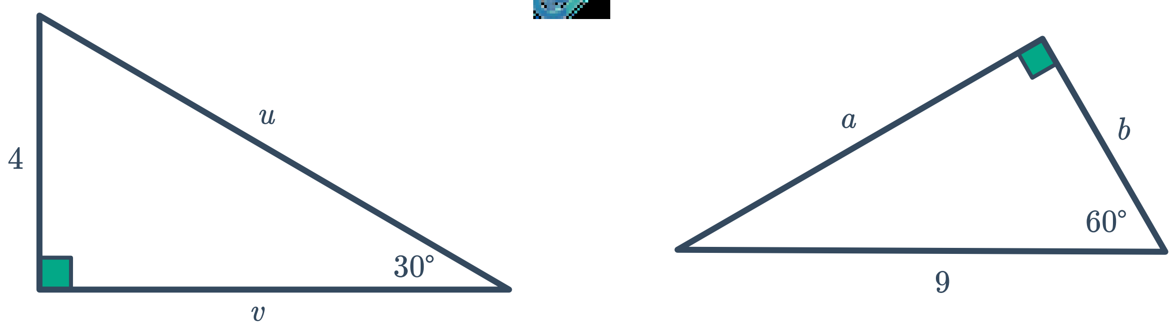
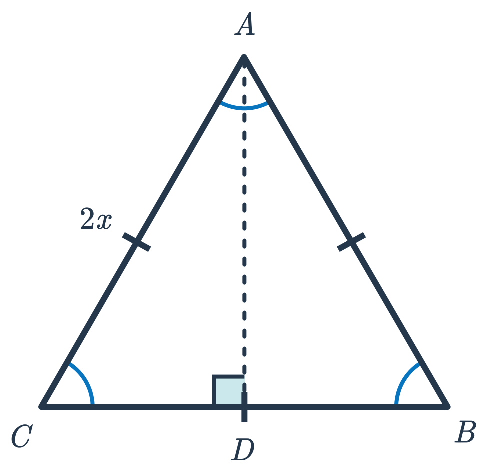


g



Applications

18 $\triangle ABC$ is an equilateral triangle with side lengths of $2x$ units. An altitude is drawn from A to meet side $\overline{BC}$ at D. This information is shown in the diagram below.



a Find the $m\angle B$.

b Given that $\overline{AD}$ bisects $\angle BAC$, find the measure of $\angle BAD$.

c Given that $\overline{AD}$ bisects $\overline{BC}$, find the length of $\overline{BD}$.

d Determine the length of the altitude $\overline{AD}$ in terms of x.

e State whether the following statements are true or false. If false, correct the statement so it is true.

- **i** The longer leg of a 30°- 60°- 90° triangle is twice the length of the shorter leg.
- **ii** The shorter leg of a 30°- 60°- 90° triangle is opposite the angle measuring 30°.
- **iii** The hypotenuse of a 30°- 60°- 90° triangle is $\sqrt{3}$ times the length of the longer leg.
- **iv** The ratio of sides in a 30°- 60°- 90° triangle is $1 : \sqrt{3} : 2$.

19 Consider the figure shown:

a Find the exact length of the following:

i p **ii** q **iii** r **iv** s

b Now, find the exact perimeter of the quadrilateral.